Activity Encircle the number which matches the word:

$4\overline{)20}$ quotient 5	$6\overline{)42}$ quotient 7	$7\overline{)56}$ quotient 8
divisor	dividend	quotient

Example 2: Solve 55 ÷ 5

$$
\begin{array}{r}
11 \\
5\overline{)55} \\
-5 \\
\hline
05 \\
-5 \\
\hline
0
\end{array}
$$

Step 1: First divide the number given in tens place by 5
5 ÷ 5 = 1. Write 1 as quotient in the tens place.
Subtract 5 from 5 we get 0.

Step 2: Then write down 5 one from dividend.
Divide the number at ones place by 5.
5 ÷ 5 = 1 write 1 as quotient in ones place.
Therefore quotient is 11 and remainder 0

EXERCISE 23

(A) Solve the following:

(1) 56 ÷ 8 = ☐ (2) 32 ÷ 4 = ☐

(3) 80 ÷ 10 = ☐ (4) 54 ÷ 6 = ☐

(5) 24 ÷ 8 = ☐ (6) 45 ÷ 5 = ☐

(7) 72 ÷ 6 = ☐ (8) 85 ÷ 5 = ☐

(9) 76 ÷ 4 = ☐ (10) 75 ÷ 5 = ☐

(B) Divide:

(1) 15 by 3

(2) 28 by 4

(3) 64 by 8

(4) 60 by 6

(5) 90 by 10

(6) 56 by 7

(7) 60 by 4

(8) 99 by 3

(9) 65 by 5

(10) 77 by 7

(11) 91 by 7

(12) 50 by 2

(C) Write the term which matches the circled number:

(1) $3\overline{)15}$ = 5 — circled: 15 — dividend

(2) $8\overline{)16}$ = 2 — circled: 2

(3) $7\overline{)28}$ = 4 — circled: 7

(4) $7\overline{)63}$ = 9 — circled: 9

(5) $8\overline{)32}$ = 4 — circled: 8

(6) $7\overline{)49}$ = 7 — circled: 49

Apply mental mathematical strategies to divide numbers up to the table of 10

Example: Saima has 24 flowers. She used 8 flowers to make a garland. How many garlands will be made with all flowers?

Solution: $24 \div 8$

First recall the table of 8, then solve

$24 \div 8 = 3$

EXERCISE 24

1. A teacher shared 28 flash cards among 4 groups of students. How many flash cards did each group will be received?
2. Ahsan has 45 tickets of charity show for sale equally in five classes. How much he sold in each class?
3. Ismail pays Rs 63 for 9 times rent of bicycle. How much did he pay for one time?
4. Raza wrote 100 pages in 5 days. How many pages he wrote daily?
5. A class teacher checks 49 copies in seven days. How many copies he checked in a day?
6. Samina used 15 litre of milk to make 5 buckets of milk rose. How much milk she used for one bucket?
7. A shopkeeper sold 81 toffees equally to 9 students. How many each student got?

Solve real life problems involving division of 2-digit numbers by 1-digit numbers

Example: **5** school vans carry total **70** students. Each van carry equal number of students. How many students are there in each van?

$$
\begin{array}{r|l}
 & 14 \\
\hline
5 & 70 \\
 & -5 \\
\hline
 & 20 \\
 & -20 \\
\hline
 & 00
\end{array}
$$

Number of students in each van **= 14**.

EXERCISE 25

(1) If one biscuit packet contains **4** biscuits, how many packets of biscuits can be made with **80** biscuits?

(2) There are **96** sweets. Each box can hold **6** sweets. How many boxes are required?

(3) **36** students are divided into volley ball team of **12** each. How many volley ball teams are made?

(4) Sana has **42** crayons. She can put **6** in one box. How many boxes will be needed?

(5) Sana purchased **6** kg salt in **Rs 84**. Find the price of **1** kg.

Unit 3

FRACTIONS

3.1 COMMON FRACTIONS

Express the fractions in figures and vice versa

Starter: The teacher will hold up a piece of circular paper and explain that this a whole.

Then the teacher will cut the circle into 4 equal parts. One part will be given to a child, on the board the teacher will write $\frac{1}{4}$ This means that one out of the four parts has been given to the child. Next the teacher will call out another child and give another piece, and then write $\frac{2}{4}$ on the board. This means two out of the for parts have been given. This will continue until all four parts have been distributed and explained.

The teacher will explain that these numbers: $\frac{1}{4}$, $\frac{2}{4}$, $\frac{3}{4}$ are called Common fractions.

In a common fraction the top number is called the numerator and the bottom number is called the denominator. For example in $\frac{3}{4}$, 3 is the numerator and 4 is the denominator. The teacher could do a few such exercises to help reinforce this concept,

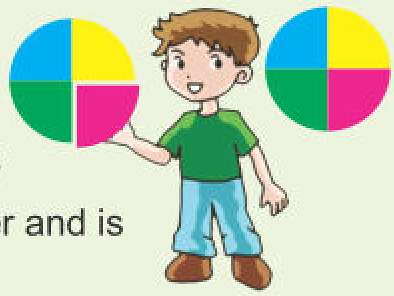
We have learnt about fractions.

A fraction is equal parts of a whole.

Some examples of fractions are $\frac{1}{2}$, $\frac{2}{3}$, $\frac{3}{4}$, etc.

We get fractions by dividing different objects in equal parts.

Suppose a circular piece of paper is cut into four equal parts. One out of 4 parts is given to a child. This piece is one-fourth of the circular piece of paper and is written as $\frac{1}{4}$. It is read as "1 by 4."



The remaining three pieces out of 4 are written as $\frac{3}{4}$. It is read as 3 by 4 or three-fourth.

These numbers $\frac{1}{4}$, $\frac{3}{4}$ and $\frac{2}{3}$ are called **Common fractions.**

In common fraction $\frac{3}{4}$, 3 is called the **numerator** and 4 is the **denominator.**

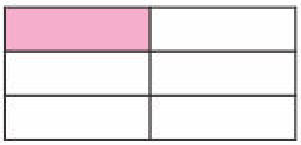
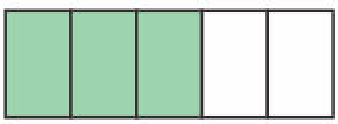
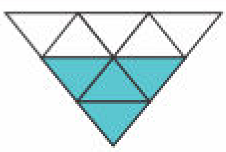
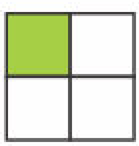
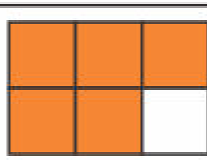
Upper number of the fraction is called Numerator $\frac{3}{4}$ Lower number of fraction is called Denominator

Examples 1:

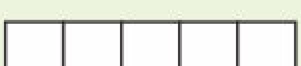
1. In $\frac{2}{9}$ Numerator is 2 and Denominator is 9.
2. In $\frac{5}{12}$ Numerator is 5 and Denominator is 12.

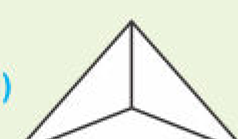
Teacher's Note Teacher should bring different real objects like biscuits and apples etc in class and cut them in equal parts to show fractions.

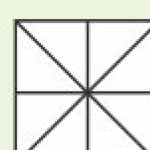
Activity 1 Express the fractions of coloured part in the following figures:

Figure	Coloured part represents	Read as
	$\frac{1}{6}$	one-sixth
	$\frac{3}{5}$	three-fifth
		
		
		

Activity 2 Show the fractions by colouring the following figures:

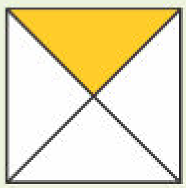
(i)  $\frac{2}{5}$

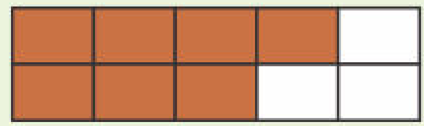
(ii)  $\frac{1}{3}$

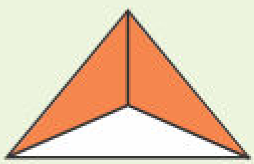
(iii)  $\frac{3}{8}$

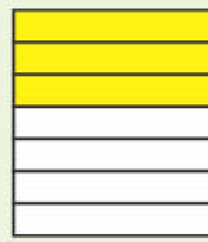
EXERCISE 26

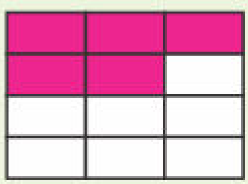
(A) Express the fractions of coloured parts of the given figures numerically.

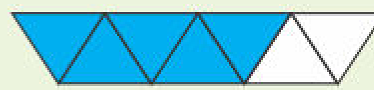
(1) 

(2) 

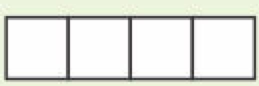
(3) 

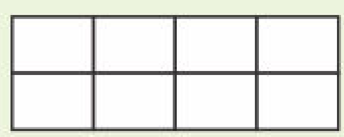
(4) 

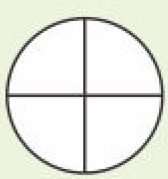
(5) 

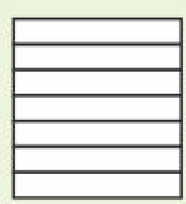
(6) 

(B) Colour the following figures to express given fractions:

(1)  $\frac{3}{4}$

(2)  $\frac{5}{8}$

(3)  $\frac{1}{4}$

(4)  $\frac{6}{7}$

(C) Write the fractions expressed by:

(1) Numerator = 4 , Denominator = 11 , fraction =

(2) Numerator = 7 , Denominator = 8 , fraction =

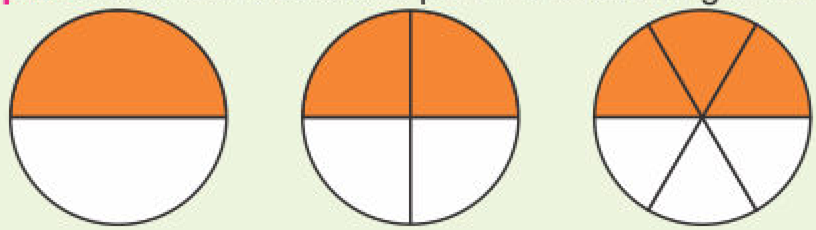
3.2 EQUIVALENT FRACTIONS

Identify equivalent fractions from the given figures.

Starter: The teacher will draw the given shaded figures on the board and explain that the coloured part in each figure represents the following fractions:

$\frac{1}{2}$, $\frac{2}{4}$, $\frac{3}{6}$

Example: Observe the coloured part in the following three figures.



The coloured part in each figure represents the following fractions:

$\frac{1}{2}$ $\frac{2}{4}$ $\frac{3}{6}$

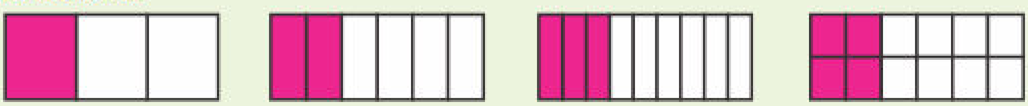
It is also observed that:

- The coloured part shown in all the three circles are of same size
- So, $\frac{1}{2}$ represents same part (half circle) as $\frac{2}{4}$ and $\frac{3}{6}$.
- Mathematically we write it as: $\frac{1}{2} = \frac{2}{4} = \frac{3}{6}$
- So, $\frac{1}{2}$, $\frac{2}{4}$, $\frac{3}{6}$ are equivalent fractions.

Write three equivalent fractions for a given fraction

Example 1: Write three equivalent fractions for a given fraction $\frac{1}{3}$

Solution:



$\frac{1}{3}$ $\frac{2}{6}$ $\frac{3}{9}$ $\frac{4}{12}$

Teacher's Note Teacher should explain the term numerator and denominator. Also explain the equivalent fraction by figure and using paper strips in class.

Here $\frac{1}{3}$, $\frac{2}{6}$, $\frac{3}{9}$ and $\frac{4}{12}$ represent same size of shape,

therefore $\frac{1}{3} = \frac{2}{6} = \frac{3}{9} = \frac{4}{12}$

Thus $\frac{2}{6}$, $\frac{3}{9}$ and $\frac{4}{12}$ are three equivalent fractions of the given fraction $\frac{1}{3}$.

Look at the following:

$$\frac{1}{3} = \frac{1 \times 2}{3 \times 2} = \frac{2}{6}, \frac{1 \times 3}{3 \times 3} = \frac{3}{9}, \frac{1 \times 4}{3 \times 4} = \frac{4}{12}$$

This explanation shows that:

> If we multiply the numerator and denominator of a fraction by the same number (other than zero), we get an equivalent fraction.

Example 2:

Find three equivalent fractions for $\frac{1}{2}$ and $\frac{2}{5}$.

Solution

$$\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \frac{4}{8}$$

Explanation

$$\frac{1 \times 2}{2 \times 2} = \frac{2}{4}, \frac{1 \times 3}{2 \times 3} = \frac{3}{6}, \frac{1 \times 4}{2 \times 4} = \frac{4}{8}$$

$$\frac{2}{5} = \frac{4}{10} = \frac{6}{15} = \frac{8}{20}$$

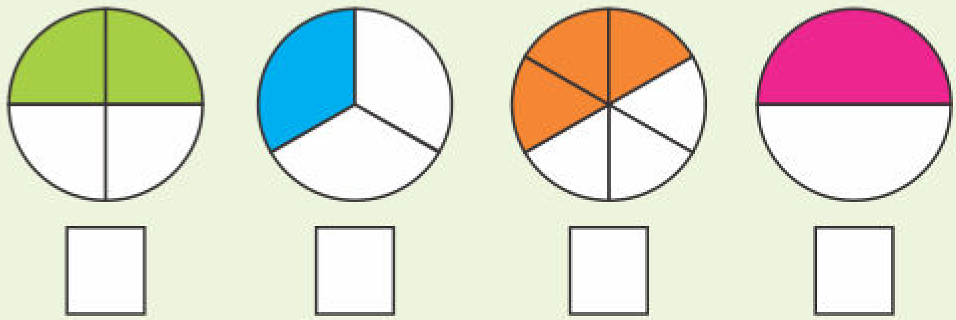
Explanation

$$\frac{2 \times 2}{5 \times 2} = \frac{4}{10}, \frac{2 \times 3}{5 \times 3} = \frac{6}{15}, \frac{2 \times 4}{5 \times 4} = \frac{8}{20}$$

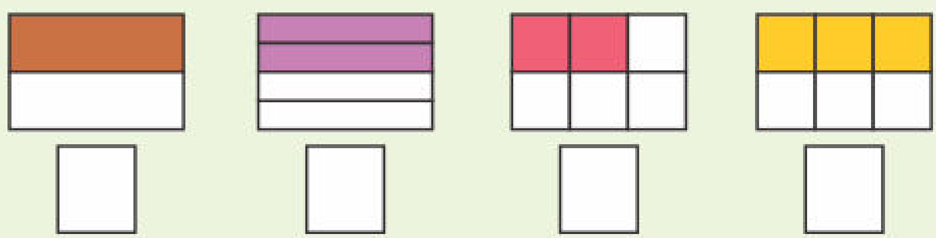
EXERCISE 27

1. Identify and write the figures which show equivalent fractions.

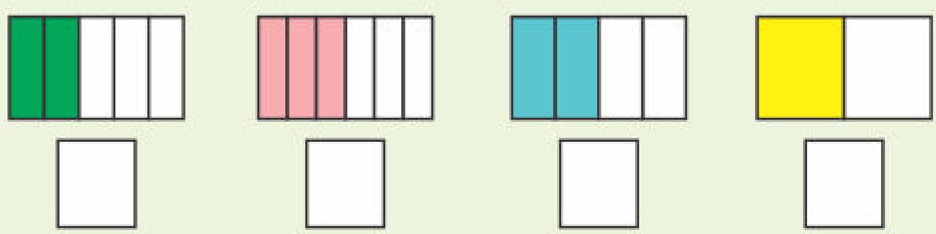
(i)



(ii)



(iii)



2. Write three equivalent fractions for the following:

(i) $\frac{1}{4}$, ___, ___, ___ (ii) $\frac{2}{3}$, ___, ___, ___

(iii) $\frac{3}{4}$, ___, ___, ___ (iv) $\frac{4}{5}$, ___, ___, ___

(v) $\frac{1}{5}$, ___, ___, ___ (vi) $\frac{2}{6}$, ___, ___, ___

3.3 PROPER AND IMPROPER FRACTIONS

Differentiate between proper and improper fractions

Starter: The teacher will write on the board a fraction in which the numerator is smaller than the denominator for example $\frac{3}{5}$.

Explain to the children that a fraction in which the numerator is smaller than the denominator is called a proper fraction.
Call some children to the board and ask them to write one example each of a proper fraction.

On the board the teacher will write a fraction in which the denominator is greater than the numerator for example such as $\frac{3}{5}$, and a fraction in which the denominator is smaller than the numerator for example such as $\frac{5}{3}$. Teacher will ask how these two fractions are different? The teacher will also encourage the children to draw the two fractions to visualize the size of two numbers.

Explain to the children that a fraction in which the denominator is smaller than the numerator is called an improper fraction.
Call some children to the board and ask them to write one example each of an improper fraction.

Unit 3 FRACTIONS

Proper Fraction:

A fraction in which numerator is smaller than the denominator is called proper fraction.

smaller

$$\frac{3}{5}$$

greater

For example $\frac{1}{4}$, $\frac{2}{5}$, $\frac{3}{7}$, $\frac{5}{6}$ and $\frac{11}{12}$ are proper fractions.

Improper Fraction:

A fraction in which numerator is greater than or equal to the denominator is called improper fraction.

greater

$$\frac{5}{3}$$

smaller

For example $\frac{4}{1}$, $\frac{5}{5}$, $\frac{6}{5}$, $\frac{5}{4}$ and $\frac{12}{11}$ are improper fractions.

$\frac{3}{1}$, $\frac{2}{1}$, $\frac{2}{2}$, $\frac{3}{3}$, $\frac{4}{4}$ are also some examples of improper fractions.

Activity **Write proper and improper fractions in the following boxes.**

$\frac{2}{9}$, $\frac{4}{3}$, $\frac{11}{7}$, $\frac{6}{1}$, $\frac{2}{5}$, $\frac{9}{9}$, $\frac{8}{1}$, $\frac{1}{3}$, $\frac{14}{13}$, $\frac{12}{15}$

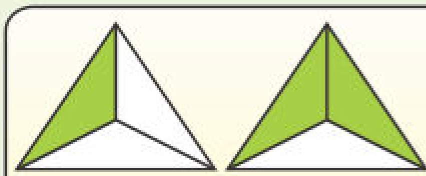
Proper Fractions	Improper Fractions

3.4 COMPARING FRACTIONS

Compare fractions with same denominators, using symbols ‘<’, ‘>’ and ‘=’

Starter: As a quick recap, the teacher will call one student at a time to the board and ask them to draw the symbols for < ,> ,= and discuss the meaning of these
And then, using the examples given explain how they can be used to compare fractions.

Let us consider the coloured parts of the following figures:



$\frac{1}{3}$ $\frac{2}{3}$

Here each figure is divided into 3 equal parts.

Fraction $\frac{1}{3}$ has **1** coloured part.

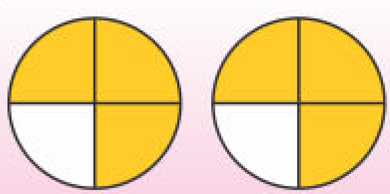
Fraction $\frac{2}{3}$ has **2** coloured parts.

As **1** is less than **2**

or in symbols **1** < **2**

So, $\frac{1}{3} < \frac{2}{3}$

Similarly, we can say $\frac{2}{3} > \frac{1}{3}$ because **2** > **1** and both fractions have same denominator.



$\frac{3}{4}$ $\frac{3}{4}$

Here, each figure is divided into **4** equal parts.

Both fractions have **3** coloured parts out of **4**.

As **3** is equal to **3**

Or **3 = 3**

So, $\frac{3}{4} = \frac{3}{4}$

If denominators of two fractions are same, the fraction with the larger numerator is the larger fraction.

Teacher’s Note Teacher should help the students to understand the comparing fractions through figures and use symbols ‘<’, ‘>’ and ‘=’.

We use symbols to compare fractions.

'<' less than	'>' greater than	'=' equal to
$\frac{1}{4} < \frac{3}{4}$	$\frac{7}{8} > \frac{5}{8}$	$\frac{2}{9} = \frac{2}{9}$
Same denominators	Same denominators	Same denominators

Activity Tick ✓ for correct and cross ✗ for incorrect.

(1) $\frac{2}{5} < \frac{3}{5}$ ✓ (2) $\frac{1}{8} > \frac{3}{8}$ ☐

(3) $\frac{11}{12} = \frac{5}{12}$ ☐ (4) $\frac{7}{13} = \frac{7}{13}$ ☐

EXERCISE 28

(A) Encircle proper fractions in the following.

(1) $\frac{2}{3}$ (2) $\frac{3}{5}$ (3) $\frac{7}{5}$ (4) $\frac{5}{1}$

(5) $\frac{4}{9}$ (6) $\frac{3}{2}$ (7) $\frac{11}{7}$ (8) $\frac{9}{9}$

(B) Fill the boxes by using symbols '<', '>' or '='.

(1) $\frac{1}{6}$ ☐ $\frac{3}{6}$ (2) $\frac{3}{5}$ ☐ $\frac{2}{5}$ (3) $\frac{2}{8}$ ☐ $\frac{4}{8}$

(4) $\frac{7}{9}$ ☐ $\frac{6}{9}$ (5) $\frac{2}{7}$ ☐ $\frac{2}{7}$ (6) $\frac{8}{10}$ ☐ $\frac{5}{10}$

(7) $\frac{3}{9}$ ☐ $\frac{4}{9}$ (8) $\frac{5}{6}$ ☐ $\frac{2}{6}$ (9) $\frac{7}{11}$ ☐ $\frac{7}{11}$

3.5 ADDITION OF FRACTIONS

Add two fractions with same denominators

Starter: Demonstrate addition of fractions through paper folding activity. For example, let children observe that an half and an half makes one; an half and one-fourth makes 3-fourth, adding one-fourth to one-fourth we get an half. The teacher will write the given example on the board and explain how to add fractions when the denominator is the same.

The teacher will call different children to the board and have them solve the sums given in the Activity

When we add the fractions with same denominators, we will add their numerators only.

Hence

Sum of two fractions having same denominators $\Rightarrow$ $\dfrac{\text{Sum of numerators}}{\text{Denominator}}$

Example: Add $\frac{3}{8} + \frac{4}{8}$

Solution: $\frac{3}{8} + \frac{4}{8} = \frac{3+4}{8} = \boxed{\frac{7}{8}}$

Activity **Solve:**

(1) $\frac{1}{7} + \frac{4}{7} = \frac{1+4}{7} = \boxed{}$

(2) $\frac{4}{9} + \frac{3}{9} = \frac{4+3}{9} = \boxed{}$

(3) $\frac{3}{11} + \frac{5}{11} = \frac{3+5}{11} = \boxed{}$